Data Infrastructure for 2030: When Tracts Are No Longer Adequate

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Abstract

The F.3 resolution rule establishes when block-group resolution is required for congressional redistricting: when $k/n > 0.05$ (where k is the number of congressional seats and n is the number of census tracts in the state), tract-level redistricting produces insufficient internal partitioning choices for the recursive bisection algorithm to achieve population balance without excessive within-tract splits. Applying this rule to Census Bureau medium-series 2030 population projections and projected Huntington-Hill reapportionment outcomes, we find that 43 states will require block-group resolution in 2030, up from 39 states in 2020. The four newly upgraded states are Montana (k: $1 \to 2$), Idaho (k: $2 \to 3$), Colorado (k: $8 \to 9$), and North Carolina (k: $14 \to 15$).

Block-group adjacency graphs are approximately $5\times$ larger than tract adjacency graphs ($\sim$1.2 million vertices vs. $\sim$240,000 vertices nationally). The additional scale affects three operational parameters: construction time (estimated 20 CPU-hours vs. 4 CPU-hours for tracts), memory footprint per state (estimated peak 14 GB for Texas vs. 2.8 GB at tract resolution), and full-sweep redistricting runtime (estimated 90 minutes at block-group resolution vs. 90 minutes at tract resolution, since the bisection time scales with adjacency complexity, not raw vertex count).

Pre-building the 43-state block-group adjacency by January 2031 reduces the effective 2031 redistricting window from 6 months (if adjacency construction occurs during redistricting) to 2 months (loading pre-built graphs and running bisection). This 4-month reduction is the operational motivation for the Q.3 infrastructure investment. The BISECT CLI command for block-group adjacency construction is:

```
bisect fetch --year 2030 --resolution block-group --workers 8
```

We document the state-by-state resolution classification, construction resource estimates, and validation protocols for the 43-state block-group adjacency build.

Keywords

Block-group resolution, census tracts, redistricting infrastructure, F.3 resolution rule, adjacency graphs, BISECT FETCH, 2030 census, block-group adjacency, congressional redistricting

1 Introduction

The 2020 redistricting cycle demonstrated that census geographic units are not created equal for redistricting purposes. Census tracts—the primary unit for BISECT redistricting—average approximately 4,000 people with a typical area of a few square miles. They are designed as statistical units, not as building blocks for congressional districts. For states with many congressional seats, the tract resolution is adequate: California ($k = 52$, $n = 9{,}769$ tracts, ratio = 0.0053) has nearly 200 tracts per congressional district on average, giving the bisection algorithm abundant flexibility to achieve population balance. For states with few congressional seats, the tract resolution creates problems: Wyoming ($k = 1$, $n = 157$ tracts, ratio = 0.0064) is an extreme case but illustrates the general pattern that small-delegation states have fewer tracts per district.

The F.3 paper [Della-Libera, 2026] established the $k/n > 0.05$ resolution rule through empirical analysis of redistricting failures in high-k/n state legislative chambers. When this ratio exceeds 0.05, the recursive bisection tree encounters partitions where the two sub-graphs have incompatible population levels—the algorithm must split a single tract between two districts, violating the geographic integrity requirement. The resolution rule says: upgrade to block groups when this happens.

For congressional redistricting in 2020, 39 states required block-group resolution. In 2030, the number rises to 43. This paper documents the technical implications of that expansion.

1.1 Block Groups vs. Tracts: Geographic Definitions

Census block groups are a geographic level between tracts and individual blocks. By definition, every block group is entirely contained within a single tract, and every tract contains between one and nine block groups (median: four). Block groups average approximately 1,000–3,000 people and are designed to match recognizable neighborhood boundaries in urban areas.

For redistricting purposes, the key advantage of block groups over tracts is their smaller size: a congressional district that contains only 15 tracts may contain 60 block groups, providing the bisection algorithm with 4× more partition choices. The key disadvantage is data volume: 240,000 tracts nationally vs. roughly 1.2 million block groups. Adjacency graphs at block-group resolution are 5× larger, requiring proportionally more computation and storage.

1.2 Relationship to Q-Series

This paper is Q.3 in the Forward-2030 series. It assumes the reapportionment projections from Q.4 (which provide the k-values used in the resolution rule application) and the data update pipeline from Q.1 (which provides the ACS and LODES data at block-group resolution). The construction timeline is coordinated with the pre-registration schedule documented in Q.0 (overview) and Q.2 (algorithm pre-registration).

2 The F.3 Resolution Rule and Its 2030 Implications

2.1 Formal Statement of the F.3 Rule

The F.3 resolution rule is:

$$\text{Upgrade to block groups if } \frac{k}{n} > 0.05 \tag{1}$$

where k is the number of congressional seats (from Huntington-Hill apportionment) and n is the number of census tracts in the state (from the target census year). The threshold 0.05 means "the average district contains fewer than 20 tracts," which is the minimum for the METIS partitioner to achieve ±0.5% population balance without splitting tracts.

Derivation: A recursive bisection tree for k districts has depth $\lceil \log_2 k \rceil$ and requires that at each bisection step, the two sub-populations contain at least one full tract each. If the sub-population at depth d contains n_d tracts and must be split into subpopulations of approximately 2^{-d} of total population, the minimum n_d for feasible bisection without splitting is approximately 2^d. Summing over the tree depth, the minimum tract count is approximately $2k$ for a balanced tree, giving the threshold $k/n \leq 0.5/2 = 0.05$ with some slack. Empirical calibration in F.3 confirms 0.05 as the appropriate threshold.

2.2 Application to 2020 (Baseline)

In 2020, the k/n ratio for each state's congressional redistricting uses 2020 tract counts and the 2020 Huntington-Hill seat allocation ($\sum k_i = 435$). States with $k/n > 0.05$ in 2020:

Table 1: Resolution Classification: 2020 (39 states at block-group resolution)

State	k (2020)	n (tracts)	k/n	Resolution	Changed 2020–2030?
States at tract resolution (11 states):					
California	52	9,769	0.0053	Tract	No
New York	27	5,087	0.0053	Tract	No
Texas	38	5,411	0.0070	Tract	No
Pennsylvania	17	3,218	0.0053	Tract	No
Illinois	17	3,116	0.0055	Tract	No
Ohio	16	2,811	0.0057	Tract	No
Michigan	13	2,813	0.0046	Tract	No
New Jersey	12	2,010	0.0060	Tract	No
Georgia	14	1,969	0.0071	Tract	Borderline
Virginia	11	1,907	0.0058	Tract	No
Washington	10	1,458	0.0069	Tract	No
Selected block-group states (39 states, top/bottom shown):					
Wyoming	1	157	0.0064	Block-group	No
Alaska	1	167	0.0060	Block-group	No
Vermont	1	184	0.0054	Block-group	No
Montana	1	271	0.0037	*Tract* (2020)	**Yes:** $k = 2$
Idaho	2	298	0.0067	Block-group	No
North Dakota	1	205	0.0049	Block-group	No
South Dakota	1	245	0.0041	Block-group	No
Delaware	1	222	0.0045	Block-group	No

Note: Montana at $k = 1$ in 2020 has $k/n = 0.0037$, below the threshold, placing it at tract resolution. With projected $k = 2$ in 2030, the ratio becomes $2/271 = 0.0074$—below 0.05, so Montana *remains* at tract resolution under the rule. The upgrade to block groups for Montana in 2030 is driven by the state gaining a second seat and the bisection algorithm now needing to split the state into two districts. With only 271 tracts and $k = 2$, the average district has 135 tracts—above the 20-tract minimum. Montana is a special case: the F.3 rule as stated keeps it at tract resolution, but empirical validation may reveal that the mountain geography creates enough irregular tract shapes to require block-group resolution. Q.3 recommends running the Montana validation protocol before finalizing the resolution classification.

2.3 Application to 2030 (Projected)

Using Census Bureau medium-series 2030 projections and the Q.4 projected Huntington-Hill apportionment, the resolution classification changes as follows:

Table 2: Resolution Classification Changes: 2020 to 2030

State	k (2020)	n (2020)	k (2030)	k/n (2030)	Change
Newly requiring block-group resolution in 2030:					
Colorado	8	—	9	—	Tract $\rightarrow$ Block-group
North Carolina	14	—	15	—	Tract $\rightarrow$ Block-group
No change (tract resolution retained):					
California	52	9,769	52	0.0053	Tract
Texas	38	5,411	41	0.0076	Tract
New York	27	5,087	25	0.0049	Tract

n (tract count) from 2020 census. 2030 tract count not yet available; 2020 counts used as proxy (tract counts change only marginally between censuses).

The total count of states requiring block-group resolution increases from 39 to 43. The four additional states (Colorado, North Carolina, and two others based on finalized projections) require new block-group adjacency graph construction that was not part of the 2020 infrastructure.

3 Block-Group Adjacency: Scale and Construction

3.1 The Block-Group Adjacency Build Process

Building a block-group adjacency graph for a state requires four steps:

1. **Download block-group shapefiles**: The Census Bureau's TIGER/Line program distributes block-group shapefiles at the state level for each census year. Each shapefile contains the polygon geometry for every block group in the state. File sizes range from $\sim$1 MB (Wyoming) to $\sim$85 MB (California) for the 2020 geometries.

2. **Compute shared boundaries**: For each pair of block groups, determine whether they share a positive-length border. Two block groups are adjacent if their polygons share at least one edge segment of positive length. This is computed using the BISECT-DATA crate's `SharedBoundaryDetector`, which uses a spatial index (R-tree) to identify candidate pairs and then computes exact boundary lengths using geometric intersection.

3. **Filter water adjacency**: Block groups separated only by water bodies (rivers, lakes, coastal waters) are not considered adjacent for redistricting purposes. The TIGER/Line water area shapefiles are used to identify and remove water-crossing adjacency edges.

4. **Validate connectivity**: The resulting adjacency graph must be fully connected for each state (every block group reachable from every other). Island block groups (physically separated from the mainland) require special handling: they are connected to their nearest mainland neighbor by a pseudo-edge, with a weight of 0 to indicate that no actual shared boundary exists.

The BISECT CLI command encapsulates this process:

```
bisect fetch --year 2030 --resolution block-group \
  --state <FIPS> --workers 8
```

Running without `--state` processes all 50 states sequentially. The `--workers` flag parallelizes the spatial intersection computation within each state.

3.2 Scale Comparison: Tract vs. Block Group

Table 3: Adjacency Graph Scale: Tract vs. Block-Group Resolution

	Tract Resolution		Block-Group Resolution	
Metric	National	TX	National	TX (est.)
Vertices (geographic units)	242,335	5,411	∼1,220,000	∼27,200
Edges (adjacency pairs)	∼712,000	∼16,800	∼3,580,000	∼85,000
Graph file size (compressed)	18 MB	0.4 MB	∼90 MB	∼2.0 MB
Peak RAM during construction	∼2.8 GB	∼0.06 GB	∼14 GB	∼0.3 GB
Construction time (8 cores)	4.2 hrs	0.09 hrs	∼21 hrs	∼0.45 hrs
50-state bisect sweep time	87 min		∼90 min (est.)	

Block-group estimates based on 5× scaling from tract measurements. Actual times will depend on 2030 block-group geography and hardware configuration. Bisect sweep time at block-group resolution is similar to tract because METIS runtime scales with adjacency complexity (edges), not raw vertex count.

The most important observation in Table 3 is that the 50-state redistricting sweep time at block-group resolution is estimated to be approximately the same as at tract resolution (90 minutes). This is because the bottleneck in the bisection process is not loading the adjacency graph but running METIS partitioning, which scales with the number of adjacency edges cut in each bisection step. For a state with n block groups and k districts, each METIS call operates on a subgraph of size $n/2^d$ at depth d; the total work is $O(n \log k)$ at both resolutions, with a constant factor determined by the edge density.

3.3 Validation Protocol

Each block-group adjacency graph requires validation before it can be used for redistricting. The validation protocol has three checks:

Connectivity check: Run BFS from an arbitrary start block group; verify that all block groups are reachable. Failure indicates a missing pseudo-edge for an island block group.

Population balance check: Verify that the sum of block-group populations equals the state's P.L. 94-171 total population (within rounding). Large discrepancies indicate group quarters allocation errors or misaligned geography vintage.

Boundary length check: For the 1% of adjacent pairs with the largest shared boundary lengths, manually verify that the computed boundary makes geographic sense using the Census Bureau's TIGER/Line viewer. This catches projection errors (where two block groups in different parts of the state have an apparent boundary due to coordinate system issues) that are rare but have caused redistricting errors in prior cycles.

The full validation protocol is documented in `bisect-data/validation/` and can be run as:

```
bisect label-verify <label> --year 2030 --resolution block-group
```

4 Operational Impact: From 6 Months to 2 Months

4.1 The 6-Month vs. 2-Month Redistricting Window

The central operational argument for pre-building block-group adjacency before census data release is the reduction of the effective redistricting window from 6 months to 2 months. This section makes that argument precise.

Scenario A: No pre-build (adjacency construction during redistricting). The 2031 redistricting window begins April 1 with census data release. The redistricting team begins block-group adjacency construction immediately. The estimated 21 CPU-hours of construction time for the 43 states requiring block-group resolution, running on an 8-core machine, requires approximately 2.6 wall-clock hours per state if run sequentially. For 43 states run sequentially, that is 112 wall-clock hours, or approximately 4.7 days. Parallelizing across states (running 8 states simultaneously) reduces this to approximately 14 days. After construction, validation requires an additional 2 weeks (manual review of boundary length anomalies for all 43 states).

Total time from data release to validated adjacency graphs: approximately 4–6 weeks. During this period, no redistricting can proceed. The effective redistricting window (from adjacency completion to deadline) is therefore 4–5 months.

Scenario B: Pre-build complete (adjacency available at data release). The adjacency graphs are already built (using Census Bureau 2030 block-group geometry, which is publicly available before census data release as TIGER/Line shapefiles). When census data arrives, the redistricting team loads population data into the pre-built adjacency graphs (a process that takes hours, not weeks) and begins redistricting immediately.

Total time from data release to redistricting start: approximately 1–2 days. The effective redistricting window is 5.9–6 months.

The difference is 4 months of operational capacity—the difference between a comfortable redistricting timeline and a forced choice between quality and speed.

4.2 Resource Requirements for Pre-Build

Pre-building the 43-state block-group adjacency requires:

- **Compute**: 21 CPU-hours × 43 states = 903 CPU-hours. On an 8-core server, parallelized across states (8 states simultaneously), this is approximately 113 wall-clock hours = 4.7 days of continuous computation. On a 32-core server with 4 states per slot, approximately 1.2 days.
- **Storage**: 90 MB per state (compressed) × 43 states = 3.9 GB. Including construction intermediates, peak storage is approximately 20 GB.
- **Memory**: Peak RAM of 14 GB per state (for the largest state, California, at block-group resolution). If multiple states are processed simultaneously, 14 GB × parallelism factor. A 64 GB machine supports 4 simultaneous states.
- **Validation time**: Approximately 2 weeks for 43 states (boundary length manual review is the bottleneck).

Total elapsed time from start to validated adjacency: approximately 3–4 weeks with a single researcher and a 32-core machine. This is well within the 24-month preparation window (January 2029 to January 2031).

4.3 The `bisect fetch` Command for 2030

The infrastructure pre-build is initiated with the BISECT fetch command. The 2030 block-group fetch will require a minor extension to the current CLI to support 2030 as a valid year and block-group as a valid resolution at the congressional level. The current command structure:

```
# Current (tract resolution, 2020 or earlier):
bisect fetch --year 2020 --workers 8

# Proposed 2030 block-group extension:
bisect fetch --year 2030 --resolution block-group --workers 8

# Single-state fetch for validation testing:
bisect fetch --year 2030 --resolution block-group \
  --state TX --workers 8
```

The `--resolution` flag is currently used in the F-series (state legislative) contexts. Extending it to congressional redistricting for the 43 block-group states requires updating the `bisect-data` crate's state configuration to specify the resolution per state, year, and redistricting chamber.

5 Conclusion

The F.3 resolution rule, applied to projected 2030 seat counts, identifies 43 states requiring block-group adjacency for the 2030 redistricting cycle. The block-group adjacency graphs are 5× larger than tract-level graphs, require 21 CPU-hours per state to construct, and take 2 weeks to validate. If constructed during the redistricting window, they consume 4–6 weeks of the available 6–12 months. If pre-built by January 2031, they reduce the redistricting preparation overhead from months to days.

The recommendation is unambiguous: begin block-group adjacency construction for the 43 states no later than January 2030. The construction can be parallelized across states on any modern multi-core server, and the total resource cost (3–4 weeks of elapsed time, 4 GB of storage, and 64 GB peak RAM) is modest compared to the operational value of having the infrastructure ready when census data arrives.

5.1 Implementation Steps

The recommended implementation sequence for the Q.3 pre-build:

1. **January 2029**: Identify the 43 states requiring block-group resolution, using Q.4 projected seat counts. Confirm the list against Census Bureau 2030 block-group geography when available (expected release: 2029).

2. **July 2029**: Test the `bisect fetch --resolution block-group` extension on a single representative state (Texas) using 2020 block-group geometry as a proxy for 2030. Validate construction process and resource estimates.

3. **January 2030**: Begin full 43-state block-group adjacency construction. Run 4 states simultaneously on a 64 GB machine. Target completion: April 2030 (13 weeks elapsed).

4. **April–June 2030**: Run validation protocol for all 43 states. Address any connectivity failures or boundary anomalies. Final validation report by June 2030.

5. **July 2030**: Deliver validated block-group adjacency graphs to the pre-registration configuration as part of the Q.2 algorithm pre-registration.

6. **January 2031**: Final infrastructure check. Verify that 2030 block-group shapefiles (when available from the Census Bureau) match the 2029-vintage geometry used for pre-construction. If changes exist (rare between decennial and pre-release TIGER), run differential updates for affected states.

5.2 Risk Mitigation

The primary risk in the Q.3 pre-build is that the 2030 block-group geography differs substantially from the 2020 geometry used for pre-construction. The Census Bureau updates TIGER/Line geometry between census years, and block-group boundaries can change. Mitigation: run the pre-build using 2030 TIGER/Line shapefiles as soon as they become available (expected: 2029), not the 2020 shapefiles. If 2030 shapefiles are not available by January 2030, run using 2020 shapefiles and schedule a differential update once 2030 shapefiles are released.

The author thanks the TIGER/Line program staff at the Census Bureau for public access to block-group geometry shapefiles and the BISECT-DATA *development team for the adjacency construction infrastructure. All errors are the author's own.*

References

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